Machine Learning
Semester 2026-III
Workshop No. 6 — Medical Domain Adaptation & Clinical Integration (Part C)

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Welcome to the sixth and final workshop of the *Machine Learning* course project!

Your team has spent five workshops developing machine learning techniques in your assigned domain. This final workshop represents a crucial transition: applying and adapting your learned models to **medical imaging and healthcare applications**. This is where you integrate your techniques, assess their clinical relevance, and reflect on what real-world deployment in healthcare would require.

Medical AI systems operate under stringent constraints: patient privacy, regulatory requirements (FDA approval, ISO standards), explainability demands from clinicians, and ethical considerations around fairness and liability. Your models must not only perform well on held-out test sets but demonstrate that they are safe, interpretable, and clinically meaningful.

The central questions for this workshop are: **How do your learned techniques transfer to medical applications? Can you explain your model's decisions to clinicians? What regulatory and ethical challenges must be addressed before clinical deployment?**

General Workshop Definition: Domain transfer and medical adaptation involve taking machine learning models and techniques developed in one domain and applying them to medical imaging, patient monitoring, or clinical decision support tasks. This requires: (1) identifying transferable components of your learned models, (2) adapting architectures and training procedures to handle medical data characteristics (resolution, modality, annotation quality), (3) implementing medical-specific evaluation metrics and clinical validation proto-

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cols, (4) producing explainability analyses that clinicians can understand and trust, and (5) assessing regulatory compliance and deployment readiness. In this workshop, teams apply domain transfer strategies, develop medical evaluation frameworks, produce clinical-grade interpretability analyses, and deliver a comprehensive assessment of healthcare deployment feasibility.

Workshop Scope and Objectives

- **Domain Transfer Strategy:** Identify which components of your domain-specific models transfer well to medical imaging. Design an adaptation strategy: feature reuse, fine-tuning, or architecture modification.
- **Medical Data Handling:** Address medical imaging specifics: high resolution, specialized modalities (X-ray, CT, MRI, ultrasound, endoscopy), class imbalance (rare diseases), and annotation quality variability.
- **Clinical Evaluation Metrics:** Move beyond standard ML metrics (accuracy, AUC-ROC) to clinical metrics: sensitivity/specificity at clinically relevant thresholds, positive/negative predictive value, and domain-specific measures (IoU for localization, dice score for segmentation).
- **Medical Interpretability:** Generate explanations (saliency maps, attention visualizations, feature attribution) that clinicians can understand and act upon. Validate that explanations align with known medical knowledge.
- **Ablation and Comparative Analysis:** Conduct ablations isolating which domain-transferred components are essential. Compare your adapted medical models against published medical imaging baselines.
- **Clinical Relevance and Regulatory Assessment:** Evaluate practical deployment requirements: FDA/regulatory pathways, privacy/HIPAA compliance, model interpretability for clinical workflows, and failure mode analysis in realistic clinical scenarios.

Medical AI and Domain Transfer — Background and References

1. **Transfer Learning in Medical Imaging:**
   - Raghu et al. (2019), "*Transfusion: Understanding Transfer Learning with Applications to Medical Imaging*", *arXiv:1902.07208* — critical analysis of when and why transfer learning works in medical imaging. Demonstrates that pretrained ImageNet features may not transfer as well as assumed; medical-specific pretraining often outperforms.
   - Tan et al. (2020), "*Efficacy and Limitations of Machine Learning Algorithms for Image-Based Urinalysis*", *American Journal of Clinical Pathology* — real-world case study of domain transfer: models trained on standard datasets often fail on new medical modalities and population shifts.
   - Srivastava et al. (2020), "*Deep Transfer Learning for Multiple Lesion Segmentation*", *IEEE TNNLS* — demonstrates successful domain transfer for medical segmentation across different imaging modalities.
2. **Medical Imaging Challenges and Evaluation:**

- Esteva et al. (2019), "*A Guide to Deep Learning*", *Nature Medicine* — primer on deep learning in healthcare, covering dataset bias, model validation, and clinical deployment.
- Oakden-Rayner et al. (2020), "*Hidden Stratification Causes Clinically Meaningful Failures in Machine Learning for Medical Imaging*", *npj Digital Medicine* — critical: models can achieve high aggregate AUC-ROC while failing systematically on subgroups (e.g., by hospital, scanner type, patient demographics). Essential reading.
- Gianfrancesco et al. (2018), "*Potential Biases in Machine Learning Algorithms Using Electronic Health Record Data*", *JAMA Internal Medicine* — analysis of data bias in healthcare AI. Medical datasets often encode historical disparities.

3. **Interpretability and Explainability in Medical AI:**

- Selvaraju et al. (2017), "*Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*", *ICCV 2017* — saliency maps for neural network interpretability. Clinical standard for visualizing model attention.
- Montavon et al. (2019), "*Explainable AI for Medical Imaging: An Introduction*", *arXiv:1908.02133* — overview of explainability methods for clinicians. Clinicians require *actionable* explanations, not just visualizations.
- Caruana et al. (2015), "*Intelligible Models for Classification and Regression*", *KDD 2015* — case study where simpler, interpretable models were preferred by clinicians over higher-accuracy black boxes.

4. **Regulatory and Ethical Frameworks:**

- FDA (2021), "*Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device*", *FDA White Paper* — regulatory requirements for AI/ML deployment in healthcare. Essential reading for deployment section.
- Selbst & Barocas (2019), "*The Fairness and Accuracy Tradeoff*", *arXiv:1806.06500* — fairness in algorithmic decision-making, with healthcare examples. Medical AI must not systematically harm subgroups.
- Rajkomar et al. (2018), "*Scalable and accurate deep learning for electronic health records*", *npj Digital Medicine* — case study: Google's EHR model achieved high performance but revealed dataset bias; required significant fairness work before deployment.

Required Artifacts and Methodology

1. **Domain Transfer Analysis:**

- Document which components of your domain-specific models (from W1–W5) are applicable to medical imaging.

- Design a transfer strategy: (a) feature reuse (freeze early layers, fine-tune final layers), (b) architecture modification (e.g., adapting input dimensions for medical image resolution), or (c) hybrid approach.
- Implement the transfer: adapt your best model(s) from W2–W5 to work with medical imaging data. If you lack access to medical imaging datasets, use publicly available medical datasets (ChestX-ray14, ISIC Skin Lesion, PathMNIST, etc.).
- Report: which transfer strategies worked? Which failed? Why?

2. **Medical Data Handling and Preprocessing:**

- Address medical imaging specifics: resolution normalization, modality-specific preprocessing (X-ray contrast stretching, CT windowing, ultrasound speckle reduction), and quality variability across images.
- Document how class imbalance (rare diseases) is handled: class weighting, sampling strategies, cost-sensitive losses, or synthetic data generation (from your W5 GAN work).
- Implement medical-specific augmentation: realistic transformations appropriate to the modality (not generic augmentations that break medical meaningfulness).

3. **Medical Evaluation Framework:**

- Define clinical metrics relevant to the task. Examples:
  - Classification (disease detection): sensitivity, specificity, PPV, NPV, AUC-ROC at clinically-relevant operating points
  - Localization (lesion detection): mean IoU, CorLoc@0.5, localization accuracy at radiologist agreement threshold
  - Segmentation: Dice coefficient, Hausdorff distance, intersection-over-union per anatomical region
  - Risk stratification: calibration error, net benefit analysis, decision curve analysis
- Compare against published medical imaging benchmarks (cite 2–3 relevant papers with baseline numbers).
- Conduct threshold analysis: report sensitivity/specificity across a range of decision thresholds. Clinicians must choose the operating point that fits their risk tolerance.

4. **Medical Interpretability Analysis:**

- Generate visual explanations for your medical AI model using Grad-CAM, attention maps, or similar techniques. For every prediction, show: input image, model output, and explanation (saliency map or feature attribution).
- Validate explanations: do they align with known medical knowledge? Do saliency maps highlight clinically-relevant anatomy? Conduct a qualitative analysis on at least 10–15 test cases.

- Document failure modes: find cases where explanations are unreliable or contradict medical knowledge. This is as important as finding successes.
- If possible, conduct expert review: have a domain expert (radiologist, pathologist, clinician) evaluate whether your model's explanations are clinically meaningful.

5. **Ablation Study (Medical Context):**
   - Conduct 3–4 ablations isolating the contribution of transferred components: (1) with full domain-transferred architecture vs. training from scratch, (2) with vs. without medical-specific preprocessing, (3) with vs. without class balancing strategies.
   - Report: how much of your medical performance comes from domain transfer vs. medical-specific adaptations?

6. **Clinical Relevance and Deployment Assessment (Minimum 600 words):**

   Write a structured assessment of clinical deployment feasibility:
   - *Clinical Problem and Stakeholders:* What clinical problem does your system solve? Which clinicians (radiologists, pathologists, primary care) would use it? How would it fit into clinical workflows?
   - *Performance Significance:* Is your model's performance clinically meaningful? Compare against radiologist performance, published benchmarks, and clinical requirements (e.g., FDA approval thresholds). Cite 2–3 relevant papers.
   - *Failure Mode Analysis:* Identify 5 common failure modes from your test set. For each, provide a concrete example and explain: Would this failure harm patients? How would a clinician catch this error?
   - *Regulatory Pathway:* What regulatory approval (FDA 510(k), PMA, lab-developed test, etc.) would be required? Discuss key obstacles.
   - *Privacy and Compliance:* Discuss HIPAA compliance, data governance, and patient consent requirements. How would you handle sensitive medical data?
   - *Fairness and Bias:* Analyze whether your model performs equally across subgroups (by age, gender, race, hospital, scanner type, disease severity). Where are disparities? How would you address them?
   - *Explainability for Clinicians:* Can radiologists understand and trust your model's decisions? Would explanations be actionable in clinical practice?
   - *Deployment Challenges:* What engineering and organizational changes would be needed? Model monitoring, retraining, performance drift detection, and user training.

7. **Comprehensive Comparison Table (Medical Context):**
   - Create a master results table comparing: all methods from W1–W6, published medical imaging baselines, radiologist performance (if available), and your domain-transferred models.

- Include clinical metrics (sensitivity, specificity, PPV, NPV) at multiple thresholds.
- Clearly highlight: best-performing method, methods that transfer well to medical domain, and methods that fail on medical data.

Final Report Structure

The final integrated report must cover:

1. *Abstract*: Problem domain (original) + medical adaptation task, transfer strategy, key results in medical domain, conclusions about clinical feasibility
2. *Introduction*: Your original domain, why medical AI is important, how your techniques could help healthcare
3. *Background*: Medical imaging modalities, clinical workflows, regulatory landscape
4. *Baseline Methods*: W1 results in original domain, published medical imaging benchmarks
5. *Domain-Specific Models*: W2–W5 results in original domain (brief)
6. *Medical Domain Transfer*: Transfer strategy, adaptation process, medical results
7. *Medical Evaluation*: Clinical metrics, comparison against baselines and radiologist performance
8. *Interpretability Analysis*: Saliency maps, explanation validation, failure mode analysis
9. *Ablation Studies*: What components transfer? What's specific to medical adaptation?
10. *Clinical Deployment Assessment*: Regulatory, privacy, fairness, explainability, operational challenges (minimum 600 words)
11. *Conclusions and Future Work*: Summary of transfer success, clinical implications, limitations, path to real-world deployment
12. *References*: Minimum 25 citations (including medical imaging, healthcare AI, regulatory, and fairness papers)

Deliverables and Documentation

1. **Final Medical Adaptation Report (`PDF`):** Following structure above. Recommended 20–30 pages (excluding references).
2. **Code:**
   - `src/transfer.py` — domain transfer implementation

- `src/medical_preprocessing.py` — medical-specific data handling
- `src/medical_metrics.py` — clinical evaluation metrics
- `src/interpretability.py` — Grad-CAM, saliency maps, explanation generation
- `src/evaluate_medical.py` — comprehensive medical evaluation
- Updated `README.md` with medical dataset links and reproduction instructions

3. **Repository:** Final state tagged `v1.0-medical-final`, all medical model checkpoints, all explanations/visualizations

Submission Requirements

- The `PDF` report must follow naming convention: `groupN_medical_final_report.pdf`
- Submit through the course platform by the specified deadline
- Include `GitHub` repository link with `v1.0-medical-final` tag
- All documentation in English, following professional scientific writing standards
- Clinical deployment assessment section must be minimum 600 words
- All reported metrics must be reproducible from submitted checkpoints and code

Grading Note: 40% of the workshop grade will be determined by an in-class quiz administered during the course. This quiz will assess your team's understanding of domain transfer principles, medical imaging considerations, clinical evaluation metrics, interpretability methods, regulatory pathways, and ethical considerations in healthcare AI. The quiz emphasizes mastery of concepts and the ability to explain transfer strategy and deployment challenges.

Important Guidelines and Considerations

- **Transfer Learning is Not Trivial:** Your domain-specific models may not transfer well to medical imaging. Negative results are valuable: document what did NOT transfer and why. This is honest science.
- **Medical Datasets are Public:** Use publicly available medical imaging datasets if you lack domain-specific data. Examples: ChestX-ray14, ISIC Skin Lesion, PathMNIST, MedMNIST, CheXpert, Kaggle medical imaging competitions.
- **Clinical Metrics Matter:** Accuracy and AUC-ROC are insufficient for medical AI. Clinicians care about sensitivity (did we catch the disease?) and specificity (did we avoid false alarms?). Report both at clinically-relevant operating points.
- **Explainability is Non-Negotiable in Healthcare:** A model that cannot be explained is not deployable in clinical practice. Saliency maps, attention visualizations, and feature attribution are clinical requirements, not optional.
- **Bias and Fairness are Critical:** Medical datasets encode historical biases. Your model may perform worse on underrepresented groups. Audit for disparities and document mitigation strategies.
- **Regulatory Awareness:** FDA approval is complex and expensive. Understanding the regulatory landscape (510(k) vs. PMA vs. lab-developed test) is essential for any healthcare AI system.

- **Ethical Responsibility:** Healthcare AI directly impacts patient outcomes. Your deployment assessment must be honest about limitations, failure modes, and when humans (clinicians) should override the model.

Congratulations on completing this comprehensive machine learning journey. You have built models in your domain of expertise and now adapted them to healthcare—one of the most challenging, regulated, and ethically important applications of machine learning. Your domain-transfer work bridges technical achievement and clinical impact. The deployment challenges you identify are not signs of failure, but honest recognition of what healthcare systems demand. Deliver your best work, and recognize that the critical thinking you apply to these challenges is what distinguishes responsible AI practitioners from technicians. The medical AI field needs practitioners like you.